

The “Differences Are Data” Discipline — when a BST-native-vs-observed difference is a DERIVED result vs a fudge. Pinned before use (checker’s half). Count as Cal reconciles.

Keeper | 2026-07-08 | Casey introduced a new methodology (describe the BST-native SM; the differences from observed are data to derive geometrically). It’s powerful and it’s right — and it can be abused. Pin the bar now, before it’s used loosely, or “it’s just the dressing” becomes over-sell #8.

The methodology (Casey)

Describe the BST-native Standard Model (BST’s own values); the difference from observed = the dressing/conversion, to be DERIVED from BST’s own structure — not an error, not imported λ QCD.

The RISK (why this needs a bar)

Left undisciplined, “the difference is data” excuses everything: compute a BST value, it’s off, and say “the rest is the dressing” — without ever deriving the dressing. That’s identical to the trap the discrete/continuous principle almost became (“the residual is just the continuum”) until we required the residual be a *named, computed* continuum. Same discipline applies here.

THE BAR — “difference is data” is a DERIVED result ONLY IF all four hold

1. The BST-native value is FORCED — target-innocent, from the geometry (not read off the observable).
2. The difference is FORWARD-DERIVED from BST’s OWN structure — the dressing computed from confinement (Hamming(7,4,3)), the bulk-color deposits, the cascade — NOT labeled “dressing” and left open, and NOT imported λ QCD. A *computed* quantity with a mechanism.
3. The target is the SCALE-CONSISTENT observed — the {1, 20, 1028} at a common scale, NOT the scheme-fuzzy {1,20,900} (Elie 4593). Compare native to a clean target.
4. The predicted difference is SPECIFIC and checkable — BST predicts *this* dressing, not “some correction of the right size.” Specificity is what makes it a program, not an excuse.

The tiers (honest, three cases)

- (a) native FORCED + difference DERIVED (from BST structure) → full bank (the native value is derived, the dressing is derived, they sum to the scale-consistent observed). This is the goal — the “differences are data” program landing.
- (b) native FORCED + difference NAMED-BUT-NOT-YET-DERIVED → the native is a LEAD, the difference is an honest open (say: “BST-native = X, forced; the dressing to the observed is Y, not yet derived from confinement”). Honest, not a bank.
- (c) native NOT forced → nothing. A native value read off the observable, with the difference called “dressing,” is a double fudge.

Why this STRENGTHENS the program (not a brake)

“Differences are data” is a *falsifiable* program when disciplined: BST predicts a *specific* dressing (the confinement running), which experiment/lattice can check. That’s stronger than “we match the SM” — it’s “here is the BST-native spectrum, and here is the specific geometric correction that dresses it to what you measure.” The paper frame Casey named (“describe the BST SM, explain the differences from geometry”) is exactly case (a) done rigorously. The bar is what makes it physics.

Applied to today's board

- $m_s/d = 20$ (scale-consistent, clean): native candidate — is $20 = \text{rank}^2 \cdot n_C$ FORCED (positions), or a value-form? (bar #1).
- $d(\nu)\text{-native} \approx \text{constituent}$ (Grace's tell): a LEAD (case b) — $d(\nu)$ native is intriguing, but the dressing (native $\rightarrow$ MS-bar) is NOT yet derived from confinement. Honest open, not a bank.
- $m_b/d \approx 1028 = N_c \cdot g^3$: reverse-read (bar #1 fails until forward-derived).
- The dressing itself (confinement running): the un-derived step — deriving it from Hamming/bulk-color is what moves $d(\nu)\text{-native}$ from case (b) to case (a).

— Keeper, 2026-07-08. The “differences are data” bar (Casey's new methodology, disciplined): a native-vs-observed difference is DERIVED only if (1) the native value is FORCED, (2) the difference is FORWARD-DERIVED from BST's own structure (confinement/Hamming, not labeled-and-left, not λ QCD), (3) the target is scale-consistent ($\{1,20,1028\}$ not $\{1,20,900\}$), (4) the predicted difference is specific/checkable. Tiers: native+difference derived = bank; native forced + difference open = lead; native not forced = nothing. Disciplined, it's a falsifiable program and a strong paper frame; undisciplined, it's over-sell #8. Count as Cal reconciles.